01 MCU en voeding van de logica

File: sheet_01.kicad_sch

02 Ingangsbeveiliging

File: sheet_02.kicad_sch

03 Relais en hardware-uitschakeling

File: sheet_03.kicad_sch

04 ITO heater

File: sheet_04.kicad_sch

05 DAC en stroominstellingen

File: sheet_05.kicad_sch

06 Vrijgave van de Peltierkanalen

File: sheet_06.kicad_sch

07 Peltier stroomregelaar 1

POWER_PRESENT

GND

POWER_PRESENT

+24_FUSED

POWER_PRESENT

+5_AUX

POWER_PRESENT

+3V3

File: sheet_07.kicad_sch

08 Peltier stroomregelaar 2

File: sheet_08.kicad_sch

09 Peltier stroomregelaar 3

File: sheet_09.kicad_sch

POWER_PRESENT

T1_VIN

POWER_PRESENT

T2_VIN

POWER_PRESENT

T3_VIN

POWER_PRESENT

T4_VIN

10 Peltier stroomregelaar 4

File: sheet_10.kicad_sch

11 Temperatuurmeting op het glas

File: sheet_11.kicad_sch

12 Heatsink, luchtvochtigheid en ventilator

File: sheet_12.kicad_sch

Sheet: /
File: ito-controller.kicad_sch

Title: ITO controller

Size: A3	Date:	**Rev: K – PROTOTYPE**
KiCad E.D.A. 9.0.9		Id: 1/13

PROTOTYPE K — see PROTOTYPE-START.md for wiring and commissioning limits

J1
XIAO right socket 1x7 — USB down
HOT_ADC
INPUT_I_ADC
LOAD_PRESENT_N
HEATER_PWM
D0
D1
D2
D3
D4
D5
D6
SDA
SCL
TEC_ARM

J2
XIAO left socket 1x7 — USB down
XIAO_5V
GND
+3V3
BUTTON_N
5V
GND
3V3
D10
D9
D8
D7
POWER_ARM
FAN_TACH
FAN_PWM

M1
TSR 1-2450
+24_PROT
GND
IN
GND
OUT
+5_AUX

JP1
Voedingsjumper — standaard OPEN
+5_AUX
1
2
XIAO_5V

C1
10u / 10V
+3V3
1
2
GND

C2
10u / 10V
+5_AUX
1
2
GND

R1
2k2
+3V3
1
2
SDA

R2
2k2
+3V3
1
2
SCL

R3
10k
+3V3
1
2
BUTTON_N

J5
Connector 2 pins
BUTTON_N
1
2
GND

R4
0R
INPUT_IMON
1
2
IMON_RAW

R5
10k 0.1%
IMON_RAW
1
2
GND

R6
1k
IMON_RAW
1
2
INPUT_I_ADC

C3
100n
INPUT_I_ADC
1
2
GND

R7
4k7
TEC_ARM
1
2
GND

Sheet: /01 MCU en voeding van de logica/
File: sheet_01.kicad_sch
Title: 01 MCU en voeding van de logica
Size: A2
Date: 2026-09-23
Rev: K — PROTOTYPE
KiCad E.D.A. 9.0.9
Id: 2/13

J3
Connector 2 pins
+24_IN
GND

F1
10A / 125V DC 0456010.MRL
+24_IN
+24_FUSED

D1
SMBJ33CA bidirectioneel
+24_FUSED
GND

U1
TPS26630RGER
EFUSE_OK
EFUSE_EN
PROT_GATE
PROT_DRV
+24_FUSED
SS_DIV
OV_DIV
GND
PROT_VUP
PROT_ILIM
PROT_GNDN
INPUT_MON
FLT
PGTH
PGOOD
OUT
OUT
NC
NC
NC
NC
NC
EP_GND
PROT_FAULT_N
PROT_PGTH
PROT_PGOOD
+24_PROT
+24_PROT

R10
156k 0.1%
+24_FUSED
OV_DIV

R11
10k 0.1%
UV_DIV
GND

R12
223k 0.1%
+24_FUSED
OV_DIV

R13
10k 0.1%
OV_DIV
GND

R14
4.02k 0.1%
PROT_ILIM
GND

R15
10k
+3V3
PROT_FAULT_N

Q2
IRLB4430PBF
PROT_GATE
EFUSE_OK
+24_FUSED

Q3
BSS138LT1G
PROT_DRV
+24_FUSED
PROT_GATE

R16
150k 0.1%
+24_PROT
PROT_PGTH

R17
10k 0.1%
PROT_PGTH
GND

R18
1k
+3V3
PROT_PGOOD

C4
100n / 50V X7R 10%
PROT_VUP
GND

C5
470u / 50V EEUFR1H471
+24_PROT
GND

C6
1u / 50V X7R
EFUSE_OK
GND

J9
EFUSE RESET NO
PROT_GNDN
GND

U15
SN74AHC1G08DBVR
POWER_ALARM
PROT_PGOOD
GND
POWER_OK_ALRM
+3V3

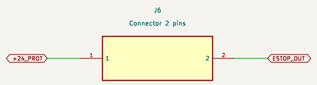
C7
100n / 10V
+3V3
GND

R19
100k
POWER_ALARM
GND

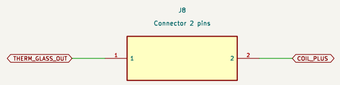
POWER_PRESENT
EFUSE_OK

Sheet: /02 Ingangsbeveiliging/
File: sheet_02.kicad_sch
Title: 02 Active input protection and relay qualification
Size: A1 | Date: | **Rev: K – PROTOTYPE**
KiCad E.D.A. 9.0.9 | Id: 3/13

PROTOTYPE K – see PROTOTYPE-START.md for wiring and commissioning limits





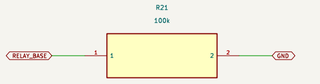




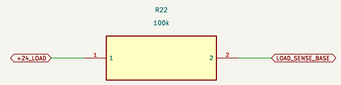


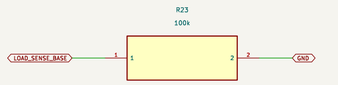


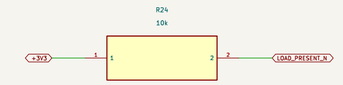


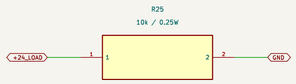


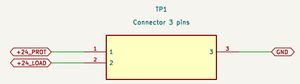






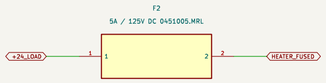


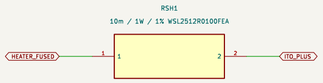




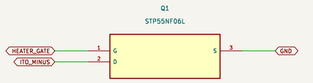
Sheet: /03 Relais en hardware-uitschakeling/		
File: sheet_03.kicad_sch		
Title: 03 Relais en hardware-uitschakeling		
Size: A2	Date: 2026-09-23	**Rev: K – PROTOTYPE**
KiCad E.D.A. 9.0.9		Id: 4/13

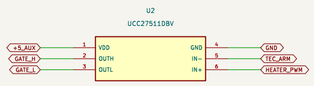
PROTOTYPE K — see PROTOTYPE-START.md for wiring and commissioning limits

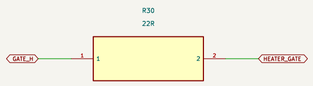


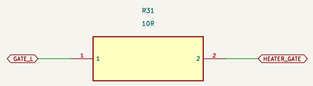


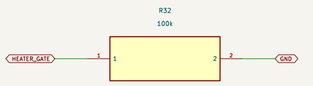


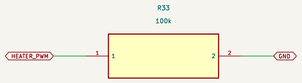


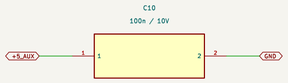


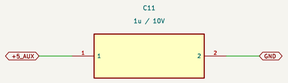


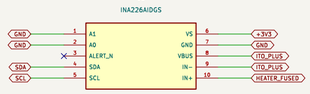


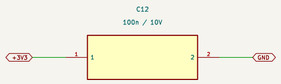












Sheet: /04 ITO heater/
File: sheet_04.kicad_sch

Title: 04 ITO heater

Size: A2	Date: 2026-09-23	**Rev: K — PROTOTYPE**
KiCad E.D.A. 9.0.9		Id: 5/13

PROTOTYPE K — see PROTOTYPE-START.md for wiring and commissioning limits

U4
MCP4728-E-UN
+3V3 SCL SDA GND
VDD SCL SDA LDAC RDY_BUSY VSS OUT_A OUT_B OUT_C OUT_D
DAC1 DAC2 DAC3 DAC4 GND

C20
100n / 10V
+3V3 GND

R42
10k 0.1%
DAC1 DIV1

R43
10k 0.1%
DIV1 GND

C21
100n / 10V
DIV1 GND

R44
10k 0.1%
DAC2 DIV2

R45
10k 0.1%
DIV2 GND

C22
100n / 10V
DIV2 GND

R46
10k 0.1%
DAC3 DIV3

R47
10k 0.1%
DIV3 GND

C23
100n / 10V
DIV3 GND

R48
10k 0.1%
DAC4 DIV4

R49
10k 0.1%
DIV4 GND

C24
100n / 10V
DIV4 GND

U5
OPA2333AID
ICTRL1 ICTRL1 DIV1 GND
OUT_A IN-_A IN+_A V-
IN+_B IN-_B OUT_B V+
DIV2 ICTRL2 ICTRL2 +3V3

C26
100n / 10V
+3V3 GND

U6
OPA2333AID
ICTRL3 ICTRL3 DIV3 GND
OUT_A IN-_A IN+_A V-
IN+_B IN-_B OUT_B V+
DIV4 ICTRL4 ICTRL4 +3V3

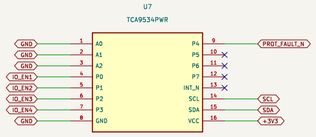
C28
100n / 10V
+3V3 GND

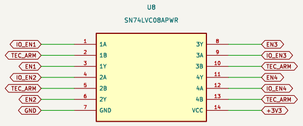
Sheet: /05 DAC en stroominstellingen/
File: sheet_05.kicad_sch

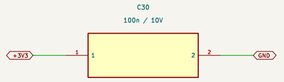
Title: 05 DAC en stroominstellingen

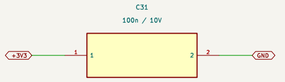
Size: A2	Date: 2026-09-23	**Rev: K — PROTOTYPE**
KiCad E.D.A. 9.0.9		Id: 6/13

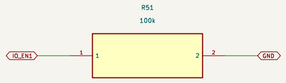
PROTOTYPE K — see PROTOTYPE-START.md for wiring and commissioning limits

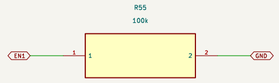




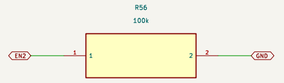


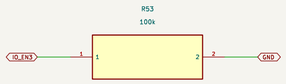


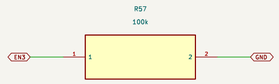


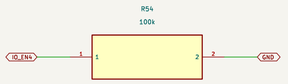


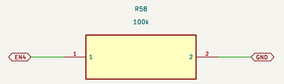












Sheet: /06 Vrijgave van de Peltierkanalen/
File: sheet_06.kicad_sch

Title: 06 Vrijgave van de Peltierkanalen

Size: A2 | Date: 2026-09-23 | **Rev: K — PROTOTYPE**
KiCad E.D.A. 9.0.9 | Id: 7/13

PROTOTYPE K — see PROTOTYPE-START.md for wiring and commissioning limits

U11
LT8613EUDE
GND 1 SYNC
T1_SS 2 TR_SS
T1_RT 3 RT
EN1 4 EN_UV
T1_VIN 5 VIN
T1_VIN 6 VIN
T1_VIN 7 VIN
GND 8 PGND
GND 9 PGND
GND 10 PGND
GND 11 GND
GND 12 GND
GND 13 GND
GND 14 GND
T1_SW 15 SW
16 SW T1_SW
17 SW T1_SW
18 SW T1_SW
19 SW T1_SW
20 BST T1_BST
21 INTVCC T1_INTVCC
22 BIAS T1_VREG
23 PG
24 FB T1_FB
25 ISP T1_VREG
26 ISN T1_OUT
27 IMON T1_IMON
28 ICTRL ICTRL
29 EP_GND GND

F11
2.5A / 125V DC 045102.5MRL
+24_LOAD 1 1 2 2 T1_VIN

C101
10u / 50V X7R
T1_VIN 1 1 2 2 GND

C102
10u / 50V X7R
T1_VIN 1 1 2 2 GND

C103
1u / 50V X7R
T1_VIN 1 1 2 2 GND

C104
1u / 10V
T1_INTVCC 1 1 2 2 GND

C105
100n / 10V
T1_BST 1 1 2 2 T1_SW

C106
10n / 10V
T1_SS 1 1 2 2 GND

R101
88k7 1%
T1_RT 1 1 2 2 GND

L1
10uH IHLP6767GZEK100M11
T1_SW 1 1 2 2 T1_VREG

C107
47u / 25V X7R C2220C476M3R2C7186
POWER_PRESENT
T1_VREG 1 1 2 2 GND

C108
47u / 25V X7R C2220C476M3R2C7186
T1_VREG 1 1 2 2 GND

R51
25m / 1W / 1% WSL2512R0250FEA
T1_VREG 1 1 2 2 T1_OUT

R102
1M13 1%
T1_OUT 1 1 2 2 T1_FB

R103
100k 1%
T1_FB 1 1 2 2 GND

C109
10p C0G
T1_OUT 1 1 2 2 T1_FB

J11
Connector 2 pins
T1_OUT 1 1 2 2 GND

TP11
Connector 2 pins
T1_IMON 1 1 2 2 GND

Sheet: /07 Peltier stroomregelaar 1/
File: sheet_07.kicad_sch
Title: 07 Peltier stroomregelaar 1
Size: A2 | Date: 2026-09-23 | **Rev: K — PROTOTYPE**
KiCad E.D.A. 9.0.9 | Id: 8/13

PROTOTYPE K — see PROTOTYPE-START.md for wiring and commissioning limits

U12
LT8613EUDE
GND 1 SYNC
T2_SS 2 TR_SS
T2_RT 3 RT
EN2 4 EN_UV
T2_VIN 5 VIN
T2_VIN 6 VIN
T2_VIN 7 VIN
GND 8 PGND
GND 9 PGND
GND 10 PGND
GND 11 GND
GND 12 GND
GND 13 GND
GND 14 GND
T2_SW 15 SW
16 SW T2_SW
17 SW T2_SW
18 SW T2_SW
19 SW T2_SW
20 BST T2_BST
21 INTVCC T2_INTVCC
22 BIAS T2_VREG
23 PG
24 FB T2_FB
25 ISP T2_VREG
26 ISN T2_OUT
27 IMON T2_IMON
28 ICTRL ICTRL2
29 EP_GND GND

F12
2.5A / 125V DC 045102.5MRL
+24_LOAD 1 1 2 2 T2_VIN

C201
10u / 50V X7R
T2_VIN 1 1 2 2 GND

C202
10u / 50V X7R
T2_VIN 1 1 2 2 GND

C203
1u / 50V X7R
T2_VIN 1 1 2 2 GND

C204
1u / 10V
T2_INTVCC 1 1 2 2 GND

C205
100n / 10V
T2_BST 1 1 2 2 T2_SW

C206
10n / 10V
T2_SS 1 1 2 2 GND

R201
88k7 1%
T2_RT 1 1 2 2 GND

L2
10uH IHLP6767GZEK100M11
T2_SW 1 1 2 2 T2_VREG

C207
47u / 25V X7R C2220C476M3R2C7186
POWER_PRESENT
T2_VREG 1 1 2 2 GND

C208
47u / 25V X7R C2220C476M3R2C7186
T2_VREG 1 1 2 2 GND

R52
25m / 1W / 1% WSL2512R0250FEA
T2_VREG 1 1 2 2 T2_OUT

R202
1M13 1%
T2_OUT 1 1 2 2 T2_FB

R203
100k 1%
T2_FB 1 1 2 2 GND

C209
10p C0G
T2_OUT 1 1 2 2 T2_FB

J12
Connector 2 pins
T2_OUT 1 1 2 2 GND

TP12
Connector 2 pins
T2_IMON 1 1 2 2 GND

Sheet: /08 Peltier stroomregelaar 2/
File: sheet_08.kicad_sch
Title: 08 Peltier stroomregelaar 2
Size: A2 | Date: 2026-09-23 | **Rev: K — PROTOTYPE**
KiCad E.D.A. 9.0.9 | Id: 9/13

PROTOTYPE K — see PROTOTYPE-START.md for wiring and commissioning limits

U13
LT8613EUDE
GND 1 SYNC
T3_SS 2 TR_SS
T3_RT 3 RT
EN3 4 EN_UV
T3_VIN 5 VIN
T3_VIN 6 VIN
T3_VIN 7 VIN
GND 8 PGND
GND 9 PGND
GND 10 PGND
GND 11 GND
GND 12 GND
GND 13 GND
GND 14 GND
T3_SW 15 SW
16 SW T3_SW
17 SW T3_SW
18 SW T3_SW
19 SW T3_SW
20 BST T3_BST
21 INTVCC T3_INTVCC
22 BIAS T3_VREG
23 PG
24 FB T3_FB
25 ISP T3_VREG
26 ISN T3_OUT
27 IMON T3_IMON
28 ICTRL ICTRL3
29 EP_GND GND

F13
2.5A / 125V DC 045102.5MRL
+24_LOAD 1 1 2 2 T3_VIN

C301
10u / 50V X7R
T3_VIN 1 1 2 2 GND

C302
10u / 50V X7R
T3_VIN 1 1 2 2 GND

C303
1u / 50V X7R
T3_VIN 1 1 2 2 GND

C304
1u / 10V
T3_INTVCC 1 1 2 2 GND

C305
100n / 10V
T3_BST 1 1 2 2 T3_SW

C306
10n / 10V
T3_SS 1 1 2 2 GND

R301
88k7 1%
T3_RT 1 1 2 2 GND

L3
10uH IHLP6767GZEK100M11
T3_SW 1 1 2 2 T3_VREG

C307
47u / 25V X7R C2220C476M3R2C7186
POWER_PRESENT
T3_VREG 1 1 2 2 GND

C308
47u / 25V X7R C2220C476M3R2C7186
T3_VREG 1 1 2 2 GND

R53
25m / 1W / 1% WSL2512R0250FEA
T3_VREG 1 1 2 2 T3_OUT

R302
1M13 1%
T3_OUT 1 1 2 2 T3_FB

R303
100k 1%
T3_FB 1 1 2 2 GND

C309
10p C0G
T3_OUT 1 1 2 2 T3_FB

J13
Connector 2 pins
T3_OUT 1 1 2 2 GND

TP13
Connector 2 pins
T3_IMON 1 1 2 2 GND

Sheet: /09 Peltier stroomregelaar 3/
File: sheet_09.kicad_sch
Title: 09 Peltier stroomregelaar 3
Size: A2 | Date: 2026-09-23 | **Rev: K — PROTOTYPE**
KiCad E.D.A. 9.0.9 | Id: 10/13

PROTOTYPE K — see PROTOTYPE-START.md for wiring and commissioning limits

U14
LT8613EUDE
GND 1 SYNC
T4_SS 2 TR_SS
T4_RT 3 RT
EN4 4 EN_UV
T4_VIN 5 VIN
T4_VIN 6 VIN
T4_VIN 7 VIN
GND 8 PGND
GND 9 PGND
GND 10 PGND
GND 11 GND
GND 12 GND
GND 13 GND
GND 14 GND
T4_SW 15 SW
16 SW T4_SW
17 SW T4_SW
18 SW T4_SW
19 SW T4_SW
20 BST T4_BST
21 INTVCC T4_INTVCC
22 BIAS T4_VREG
23 PG
24 FB T4_FB
25 ISP T4_VREG
26 ISN T4_OUT
27 IMON T4_IMON
28 ICTRL ICTRL
29 EP_GND GND

F14
2.5A / 125V DC 045102.5MRL
+24_LOAD 1 1 2 2 T4_VIN

C401
10u / 50V X7R
T4_VIN 1 1 2 2 GND

C402
10u / 50V X7R
T4_VIN 1 1 2 2 GND

C403
1u / 50V X7R
T4_VIN 1 1 2 2 GND

C404
1u / 10V
T4_INTVCC 1 1 2 2 GND

C405
100n / 10V
T4_BST 1 1 2 2 T4_SW

C406
10n / 10V
T4_SS 1 1 2 2 GND

R401
88k7 1%
T4_RT 1 1 2 2 GND

L4
10uH IHLP6767GZEK100M11
T4_SW 1 1 2 2 T4_VREG

C407
47u / 25V X7R C2220C476M3R2C7186
POWER_PRESENT
T4_VREG 1 1 2 2 GND

C408
47u / 25V X7R C2220C476M3R2C7186
T4_VREG 1 1 2 2 GND

R54
25m / 1W / 1% WSL2512R0250FEA
T4_VREG 1 1 2 2 T4_OUT

R402
1M13 1%
T4_OUT 1 1 2 2 T4_FB

R403
100k 1%
T4_FB 1 1 2 2 GND

C409
10p C0G
T4_OUT 1 1 2 2 T4_FB

J14
Connector 2 pins
T4_OUT 1 1 2 2 GND

TP14
Connector 2 pins
T4_IMON 1 1 2 2 GND

Sheet: /10 Peltier stroomregelaar 4/
File: sheet_10.kicad_sch
Title: 10 Peltier stroomregelaar 4
Size: A2 | Date: 2026-09-23 | **Rev: K — PROTOTYPE**
KiCad E.D.A. 9.0.9 | Id: 11/13

PROTOTYPE K — see PROTOTYPE-START.md for wiring and commissioning limits

U9
ADS1115IDGS
GND
GND
TEMP_ADC1
TEMP_ADC2
ADDR
ALERT_RDY
GND
AIN0
AIN1
AIN2
AIN3
VDD
SDA
SCL
TEMP_ADC3
TEMP_ADC4
+3V3
SDA
SCL

C40
100n / 10V
+3V3
GND

R61
10k 0.1%
+3V3
NTC1

R65
1k
NTC1
TEMP_ADC1

C41
100n / 10V
TEMP_ADC1
GND

J21
Connector 2 pins
NTC1
GND

R62
10k 0.1%
+3V3
NTC2

R66
1k
NTC2
TEMP_ADC2

C42
100n / 10V
TEMP_ADC2
GND

J22
Connector 2 pins
NTC2
GND

R63
10k 0.1%
+3V3
NTC3

R67
1k
NTC3
TEMP_ADC3

C43
100n / 10V
TEMP_ADC3
GND

J23
Connector 2 pins
NTC3
GND

R64
10k 0.1%
+3V3
NTC4

R68
1k
NTC4
TEMP_ADC4

C44
100n / 10V
TEMP_ADC4
GND

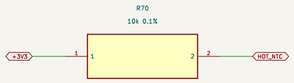
J24
Connector 2 pins
NTC4
GND

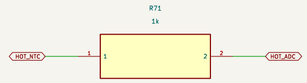
Sheet: /11 Temperatuurmeting op het glas/
File: sheet_11.kicad_sch

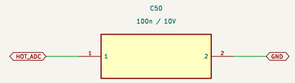
Title: 11 Temperatuurmeting op het glas

Size: A2	Date: 2026-09-23	**Rev: K — PROTOTYPE**
KiCad E.D.A. 9.0.9		Id: 12/13

PROTOTYPE K — see PROTOTYPE-START.md for wiring and commissioning limits

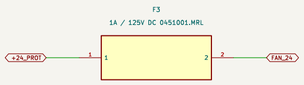


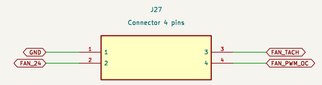


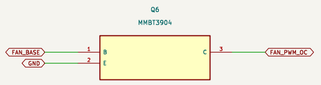






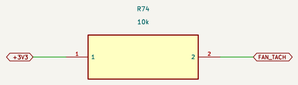












Sheet: /12 Heatsink, luchtvochtigheid en ventilator/
File: sheet_12.kicad_sch

Title: 12 Heatsink, luchtvochtigheid en ventilator

Size: A2	Date: 2026-09-23	**Rev: K — PROTOTYPE**
KiCad E.D.A. 9.0.9		Id: 13/13